



Skill Upgradation for Rural Education and Professional Education

Assignment -3 - validation.

Domain	Physical design
Under the guidance of	Veeramani R Synopsis Preethi grace
Submitted by	Bodugam naga sai Kiran
Sure pro-ed ID	saikiran-G3-25-VLSI



```
arjun123@vlsi: ~/Desktop/Openlane/OpenLane
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs$ cd picorv32
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs/picorv32$ cd src
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs/picorv32/src$ ls
picorv32.v validation_mod.tcl
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs/picorv32/src$ nano validation_mod.tcl
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs/picorv32/src$ cd ..
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs/picorv32$ cd ..
arjun123@vlsi:~/Desktop/Openlane/OpenLane/designs$ cd ..
arjun123@vlsi:~/Desktop/Openlane/OpenLane$ make mount
cd /home/arjun123/Desktop/Openlane/OpenLane && \
docker run --rm -v /home/arjun123/Desktop/Openlane/OpenLane:openlane -v /home/arjun123/Desktop/Openlane/OpenLane/designs:openlane/install -v /home/arjun123/home/arjun123 -v /home/arjun123/.ciel:/home/arjun123/.ciel --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-and64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane
1.0.2
% source designs/picorv32/src/validation_mod.tcl
===== Floorplan & Std Cell Area =====
Core Area : Width=448.04 um, Height=446.00 um, Aspect=0.9956253905901250, Area=199061.6832 um^2
Die Area : Width=459.08 um, Height=469.8 um, Aspect=1.0233510499259388, Area=215675.78399999999 um^2
Std Cell Area : 100370.012800 um^2

[✓] Floorplan dimensions and standard cell area saved to: floorplan_stdcell_report.csv
===== Design Utilization Report =====
Stage | Utilization (%)
-----|-----
Floorplan | Log file missing
Global_Placement | 52
Detailed_Placement | 51
CTS | 54
Post_CTS_Resizer | 62
Global_Routing | 62

[✓] Design utilization values saved to: design_utilization_report.csv
Stage | Log File | Total Power (W)
-----|-----
Synthesis | 2-sta.log | 2.46e-02
Global_Placement | 8-gpl_sta.log | 2.87e-02
Detailed_Placement | 11-dpl_sta.log | 2.72e-02
CTS | 13-cts_sta.log | 3.37e-02
Post_CTS_Resizer | 18-rsz_timing_sta.log | 3.31e-02
Global_Routing | 21-grt_sta.log | 3.04e-02
Signoff_STA_Max | 28-rcx_mcsta.max.log | 5.11e-02

[✓] Power values saved to: power_report.csv
Detected clock period: 0.0 ns
-----
Stage | Log File | ClkPeriod | InDelay | In(%) | OutDelay | Out(%)
-----|-----
Synthesis | 2-sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Global_Placement | 8-gpl_sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Detailed_Placement | 11-dpl_sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
CTS | 13-cts_sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Post_CTS_Resizer | 18-rsz_timing_sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Global_Routing | 21-grt_sta.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Signoff_STA_MIN | 26-rcx_mcsta.min.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
Signoff_STA_MAX | 28-rcx_mcsta.max.log | 8.00 | 2.0 | 25.00 | 2.0 | 25.00
```



arjun123@vlsi: ~/Desktop/Openlane/OpenLane

arjun123@vlsi: ~/Desktop/Openlane/OpenLane/design

CTS	13-cts_sta.log	3.37e-02
Post_CTS_Resizer	18-rsz_timing_sta.log	3.31e-02
Global_Routing	21-grt_sta.log	3.04e-02
Signoff_STA_Max	28-rcx_mcsta.max.log	5.11e-02

✓ Power values saved to: power_report.csv

Detected clock period: 8.0 ns

Stage	Log File	ClkPeriod	InDelay	In(%)	OutDelay	Out(%)
Synthesis	2-sta.log	8.00	2.0	25.00	2.0	25.00
Global_Placement	8-gpl_sta.log	8.00	2.0	25.00	2.0	25.00
Detailed_Placement	11-dpl_sta.log	8.00	2.0	25.00	2.0	25.00
CTS	13-cts_sta.log	8.00	2.0	25.00	2.0	25.00
Post_CTS_Resizer	18-rsz_timing_sta.log	8.00	2.0	25.00	2.0	25.00
Global_Routing	21-grt_sta.log	8.00	2.0	25.00	2.0	25.00
Signoff_STA_MIN	26-rcx_mcsta.min.log	8.00	2.0	25.00	2.0	25.00
Signoff_STA_MAX	28-rcx_mcsta.max.log	8.00	2.0	25.00	2.0	25.00
Signoff_STA_nom	30-rcx_mcsta.nom.log	8.00	2.0	25.00	2.0	25.00

✓ Saved CSV: in_out_delay_report.csv

Stage	Log File	Worst Setup	Worst Hold
Synthesis	2-sta.log	0.02	1.39
Global_Placement	8-gpl_sta.log	0.03	1.15
Detailed_Placement	11-dpl_sta.log	0.04	1.49
CTS	13-cts_sta.log	0.02	1.45
Global_Routing	21-grt_sta.log	0.18	2.12
Signoff_STA_MIN	26-rcx_mcsta.min.log	-5.43	0.05
Signoff_STA_MAX	28-rcx_mcsta.max.log	-5.86	0.07

✓ Worst setup/hold slack per stage saved to: worst_slack_report.csv

uncertainty

Stage: synthesis --> Setup = 0.4, Hold = 0.40

Stage: placement --> Setup = 0.4, Hold = 0.40

Stage: cts --> Setup = 0.4, Hold = 0.40

Stage: routing --> Setup = 0.4, Hold = 0.40

Stage: signoff --> Setup = NA, Hold = NA

✓ Saved uncertainties to uncertainty_report.csv

Stage	Type	Startpoint	Endpoint	Worst Slack
Synthesis	Setup	_15922_	_16079_	1.39
Synthesis	Hold	_15920_	_15920_	0.02
Global_Placement	Setup	_15922_	_16079_	1.15
Global_Placement	Hold	_15983_	_15983_	0.03
Detailed_Placement	Setup	_16114_	_15918_	1.49
Detailed_Placement	Hold	_16239_	_16336_	0.04
CTS	Setup	_16114_	_15918_	1.45
CTS	Hold	_16243_	_16340_	0.02
Global_Routing	Setup	_15922_	_16079_	2.12
Global_Routing	Hold	_16110_	_16150_	0.18
Signoff_STA_MIN	Setup	_16114_	_15918_	-5.43
Signoff_STA_MIN	Hold	_15989_	_16054_	0.05
Signoff_STA_MAX	Setup	_16114_	_15918_	-5.86
Signoff_STA_MAX	Hold	_15917_	_15917_	0.07
Signoff_STA_NOM	Setup	_16114_	_15918_	-5.64
Signoff_STA_NOM	Hold	_15989_	_16054_	0.06

✓ All reg-to-reg worst setup and hold slacks saved to: reg2reg_worst_slack.csv

% █



```

arjun123@vlsi:~/Desktop/Openlane/OpenLane$ make mount
cd /home/arjun123/Desktop/Openlane/OpenLane && \
    docker run --rm -v /home/arjun123/Desktop/Openlane/OpenLane:/openlane -v /home/arjun123/
Desktop/Openlane/OpenLane/designs:/openlane/install -v /home/arjun123:/home/arjun123 -v /home/
arjun123/.ciel:/home/arjun123/.ciel -e PDK_ROOT=/home/arjun123/.ciel -e PDK=sky130A --user
1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt
seccomp=unconfined -ti ghcr.io/the-openroad-project/
openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

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